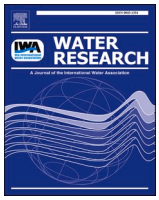
Water Research 268 (2025) 122779



Contents lists available at ScienceDirect

Water Research

journal homepage: www.elsevier.com/locate/watres





Explainable artificial intelligence for reliable water demand forecasting to increase trust in predictions

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ARTICLE INFO

Keywords:
Battle of water demand forecasting
EU AI act
Machine learning
Opaque
Transparent
Water supply system
XAI

ABSTRACT

The "EU Artificial Intelligence Act" sets a framework for the implementation of artificial intelligence (AI) in Europe. As a legal assessment reveals, AI applications in water supply systems are categorised as high-risk AI if a failure in the AI application results in a significant impact on physical infrastructure or supply reliability. The use case of water demand forecasts with AI for automatic tank operation is for example categorised as high-risk AI and must fulfil specific requirements regarding model transparency (traceability, explainability) and technical robustness (accuracy, reliability). To this end, six widely established machine learning models, including both transparent and opaque models, are applied to different datasets for daily water demand forecasting and the requirements regarding model accuracy, transparency and technical robustness are systematically evaluated for this use case. Opaque models generally achieve higher prediction accuracy compared to transparent models due to their ability to capture the complex relationship between parameters like for example weather data and water demand. However, this also makes them vulnerable to deviations and irregularities in weather forecasts and historical water demand. In contrast, transparent models rely mainly on historical water demand data for the utilised dataset and are less influenced by weather data, making them more robust against various data irregularities. In summary, both transparent and opaque models can fulfil the requirements regarding explainability but differ in their level of transparency and robustness to input errors. The choice of model depends also on the operator's preferences and the context of the application.

1. Introduction

The "Artificial Intelligence (AI) Act" was published in July 2024 to establish a legal framework for AI within the European Union (EU) (European Parliament and Council, 2024). The AI Act aims to ensure that AI systems are safe, trustworthy, and respect fundamental rights and values, especially in high-risk AI applications. Simplified, high-risk AI refers to applications that can have significant detrimental effects on the performance in the event of failures and/or lack of transparency. Among these are AI applications in critical infrastructure, which must meet special requirements regarding design, training, and modelling procedure, as well as a careful assessment of model accuracy versus explainability.

As data availability for urban water infrastructure has significantly increased over recent decades (Eggimann et al., 2017), machine learning (ML), as a subset of AI, is now frequently utilised in urban water supply systems (WSS) (Huang et al., 2021; Rapp et al., 2023). Examples of ML applications in WSS include demand and system state forecasting, leakage detection and localisation, pump and valve operations, and optimal design (Ahmed et al., 2024; Boo et al., 2024; Fu et al., 2022; Oğuz and Ertuğrul, 2023). Additionally, recently ML models are also being utilised as metamodels for hydraulic and quality simulations (Garzón et al., 2022; Haggerty et al., 2023). As WSS are part of critical infrastructure, implementing ML in accordance with the new directive, the EU AI Act, is of the greatest importance. However, despite the rapidly growing number of publications in this field, to the best of the authors' knowledge, an assessment of the impact of the AI Act on WSS and state-of-the art implementations are still missing in the literature.

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https://doi.org/10.1016/j.watres.2024.122779
Received 22 July 2024; Received in revised form 5 November 2024; Accepted 8 November 2024
Available online 9 November 2024
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As an intermediate goal for trustworthiness, research has focused on the development of Explainable AI (XAI). Generally, XAI encompasses tools and methods designed to explain and understand the predictions of ML models, distinguishing between integrated (transparency-based) and post-hoc approaches (Došilović et al., 2018). Integrated interpretability refers to low-complexity models such as Linear Regression, Decision Trees, and Rule-Based Models, often categorised as transparent models in literature. In contrast, post-hoc approaches are associated with Support Vector Regression, Random Forests, and Artificial Neural Networks, including deep learning, which are often referred to as opaque or black-box models (Belle and Papantonis, 2021). Transparent models have a limited model size and sparsity, allowing their process to be understood and recalculated by humans. Despite their limited accuracy, particularly for more complex data sets due to their simple model structure, these models remain popular in WSS operations due to their ease of explainability (Niknam et al., 2022).

In contrast, opaque models have a more complex structure, allowing them to model more intricate relationships, which generally leads to higher accuracy compared to transparent models (Huang et al., 2021). In recent years, deep learning techniques, such as Long Short-Term Memory, have gained increased popularity due to their improved prediction accuracy and their ability to capture temporal dependencies (Ahmed et al., 2024; Fu et al., 2022; Kavya et al., 2023). However, this increased accuracy often comes at the cost of reduced explainability, particularly as deep learning techniques may involve thousands of model parameters (Garzón et al., 2022; Tripathy and Mishra, 2024). As a result, opaque ML models require additional (post-hoc) explanations to interpret their results, including natural language explanations, visualisations, and other techniques, which can be either model-agnostic or model-specific (Belle and Papantonis, 2021).

Research on XAI is a fast-growing field, but to the best of the authors' knowledge, it is currently limited to the following publications related to WSS. Typically, the performance of different ML models is compared first, followed by the identification and quantification of the most important input feature of the best-performing ML model. This approach was, for instance, used by Ou et al. (2023); Rachunok et al. (2024), and Xenochristou et al. (2021) for water demand forecasting. The results showed that, although an opaque model with post-hoc explainability was utilised in each case, the most important input parameters varied between the case studies. Similar approaches have been applied to groundwater modelling (Liu et al., 2022) and water quality predictions (Nallakaruppan et al., 2024; Yang et al., 2024).

As the literature on XAI shows, different ML models are compared in terms of accuracy, but decisions are only explained for the best-performing model in each case. Given the differences in the most important input parameters between individual case studies for the same application, there is a need for comparison between different ML models in terms of explainability. This is particularly important to meet the requirements of the new directive, which calls for a careful assessment of model accuracy versus explainability. Additionally, reliability analyses of the applied ML models are necessary, such as evaluating how errors in the input data affect accuracy – an area that represents another research gap in the field of XAI for WSS.

To address these research gaps, different transparent and opaque ML models are trained for the exemplary application of short-term water demand forecasting in water demand management, using a freely available dataset. Subsequently, the applied ML models are extended with post-hoc explainability to answer specific questions such as: "*Which input features are selected, and how do changes in feature values affect the predictions?*", "*How does the model reach its decision for a specific data point?*", and "*What is the impact of deviations in the training data?*". This approach provides an in-depth understanding of the applied ML models, ensuring they meet the requirement of the new directive.

Subsequently, the aim of this work can be described as follows:

- Assessing the impact of the European Artificial Intelligence Act on urban water supply systems as critical infrastructure.
- Exemplary implementation and comparison of transparent and opaque ML models for short-term water demand forecasts (up to seven days) on a daily basis. For the selected use case of targeted water demand management (e.g., tank operation), the key requirements of technical robustness and safety and transparency are considered with a focus on model accuracy, reliability and explainability.

This work adopts an interdisciplinary approach, consisting of two parts. The first part includes a legal assessment of the "EU Artificial Intelligence Act" (European Parliament and Council, 2024) and the Ethnic Guidelines for Trustworthy AI provided by the High-Level Expert Group on Artificial Intelligence of the European Commission (European Commission, 2019). This section focuses on assessing the impact of the new directive on the WSS, with a special focus on the selected use case of water demand forecasting. The second and main part of this work provides an overview of model accuracy, a technical comparison of the explainability of transparent and opaque ML models for short-term water demand forecasts, and an uncertainty assessment to address specific questions regarding model reliability.

2. EU artificial intelligence act – impact for water supply systems

2.1. EU artificial intelligence act

Water supply systems (WSS) are part of critical infrastructure where errors and/or lack of transparency in AI systems can significantly impact both the deployer and the affected individuals. Therefore, it seems evident that the EU AI Act should be applied specifically to WSS. However, upon closer inspection, a differentiation is necessary. For critical infrastructure, the AI Act focuses on AI applications intended as safety components, fulfilling safety functions for the AI systems. This includes cases where failure or malfunction endangers the health and safety of persons or property in the management and operation of WSS (European Parliament and Council, 2024). Given the wording of the provision (based on safety components), it would be too simplistic to categorise all AI systems in WSS as high-risk.

In the meaning of the AI Act, the covered safety components of critical infrastructure, including critical digital infrastructure, are understood to be systems used to protect the physical integrity of the critical infrastructure or the health and safety of persons and property. However, these components are not essential for the system's functionality (European Parliament and Council, 2024). A failure or malfunctioning of such components might directly lead to risks to the physical integrity of critical infrastructure and, consequently, to the health and safety of people and property. In contrast, an AI system is not considered high-risk, even if mentioned in Annex III of the AI Act, if it does not pose a significant risk of harm to the health, safety, or fundamental rights of natural persons. The concept of critical infrastructure and the associated high-risk AI systems covered by the AI Act remain somewhat vague despite detailed descriptions. Therefore, it requires a case-by-case assessment for each system. This flexible regulatory approach is not unique to the AI Act and aligns with similar regulatory frameworks, characterised as a flexible system of individual conceptual characteristics (European Parliament and Council, 2008). However, this approach introduces uncertainties for the user.

2.2. Ethics guidelines for trustworthy AI

Following on from these fundamental considerations, it is necessary to take a closer look at some of the details to gain a better understanding of the implications of the legal basis. Thereby, the Ethics Guidelines for Trustworthy AI provided by the High-Level Expert Group on AI of the

European Commission (European Commission, 2019) play a decisive role in the legal interpretation of the AI Act. In the document, seven key requirements for the realisation of trustworthy AI are listed, including transparency as well as technical robustness and safety. Transparency encompasses the aspects of traceability, explainability, and communication. Technical robustness and safety are primarily based on the principles of avoiding harm and protecting people. To meet this requirement, the following properties should be implemented: resilience to attack and security, fallback plan and general safety, accuracy and reliability, and reproducibility.

This work focuses on the model-based requirements:

- Traceability (documentation of data sets, algorithms, processes, and decisions in an AI system's development)
- Explainability (human ability to understand and trace the AI decisions)
- Accuracy (providing correct predictions)
- Reliability (functioning reliably under a wide range of scenarios and input parameters, avoiding unfavourable decisions)
- Communication (open communication about the limitations of the AI system).

The remaining aspects depend heavily on the design of the integration of the AI system into operations and are therefore process-based.

According to Article 13 of the AI Act, high-risk AI systems must be designed to ensure sufficient transparency, allowing deployers to appropriately interpret and use their results. Transparency must be provided in an appropriate type and to an appropriate degree so that providers and deployers can meet their obligations as outlined. This means that the requirements must be evaluated in relation to the specific individual case, with the legal term "appropriate" used in the AI Act providing the necessary flexibility to ensure a proportionate application of the AI Act in practice. Additionally, high-risk AI systems must be designed to allow effective human oversight, including with suitable human-machine interface tools (Article 14 of the AI Act). The supervisory measures must be appropriate (with other words proportionate) to the risks, the degree of autonomy, and the context of the high-risk AI system's use. This indicates that the concept of transparency follows and adapts the risk level of the AI system.

In case of opaque models, other explainability measures (e.g., traceability, auditability, and transparent communication on the AI system's capabilities) may be required, provided the AI system respects fundamental rights. The need for explainability depends on the context and the severity of the consequences of erroneous or inaccurate outputs on human life. This implies that opaque AI systems are permissible in principle. Their permissibility, based on the risk-based approach of the AI Act, depends on weighting the positive and potential negative consequences of misuse. Whether opaque methods for water demand forecasting meet the AI Act's requirement cannot be answered generally.

Both, transparent and opaque models with post-hoc explanations can fulfil the transparency requirements of the AI Act. For opaque models, transparency can be sufficiently guaranteed by measures, such as traceability, verifiability, and transparent communication about the AI system's capabilities. The key benchmark is the severity of erroneous or inaccurate results for those affected. A general prohibition of opaque models with post-hoc explainability as inadmissible would contradict the main objectives of the AI Act, which recognises the (dynamic) opportunities of such models and are risk-based. The Ethic Guidelines explicitly endorse compromises between greater precision of AI systems and reduced explainability. For further information, a detailed legal assessment is included in the appendix.

2.3. Practical significance for water supply systems

Under the AI Act, AI applications are classified as high-risk if a failure could compromise the physical integrity of the WSS and/or the health and safety of people and property. Therefore, there are valid reasons to consider the selected use case of targeted water demand as a high-risk AI systems - especially if it could negatively impact the WSS, including causing supply shortages. Similarly, AI applications in system operations (e.g., pump and valve operations, evaluation of key performance indicators) or optimal design can also be categorised as high-risk AI systems. For high-risk AI, both transparent and opaque models are viable options, and implementation requires a balanced assessment of the accuracy and explainability of different models. Consequently, opaque models have to be extended with appropriate post-hoc explainability to ensure transparent communication of their benefits, drawbacks, and limitations.

In contrast, a system that merely supports the prediction of water consumption to automatically detect new leakages is less critical. However, the assessment ultimately depends on the AI system's exact design. If incorrect results only cause the operator to search the supply network for non-existent leakages, it does not meet the criteria for a high-risk AI system, as it is non-essential and has no expected effects on protected legal interests. The situation changes if such a system is the only criterion for assessing the (non-)existence of leakages. In that case, the system would be essential not only for maintaining the physical integrity of the WSS but also for protecting the health and safety of people and property in the event of an imminent failure.

AI applications used for leakage localisation and for evaluation of measurement parameters not utilised as decision parameters can be classified as non-critical. This also applies to AI applications that are used for administration tasks or customer service.

3. Material and methods

As discussed in the Ethical Guidelines for Trustworthy AI (European Commission, 2019), traceability is a fundamental requirement for transparency. It involves the disclosure and comprehensive documentation of all steps in model development. This includes documenting the data sets used, their pre-processing steps, the process of splitting data into training, testing, and validation sets, as well as the procedures for model selection and hyper-parameter tuning. This documentation should enable a third party to understand all decisions made throughout the AI system's development process.

3.1. Machine learning approaches

In this work, six standard ML models widely used in water demand forecasting (Kavya et al., 2023) were compared and tested. Specifically, Linear Least Squares Regression (LS), Decision Tree Regression (DT), and K-Nearest Neighbours Regression (KNN) were employed as transparent models. On the other hand, Support Vector Regression with radial kernel (SV), Random Forest Regression (RF), and Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM) were utilised as opaque models. These selected ML models cover a broad spectrum of methodologies, ranging from very simple to more complex models. For a detailed description and comprehensive analysis of model transparency, references are made to the respective literature, such as Barredo Arrieta et al. (2020) or Belle and Papantonis (2021).

3.2. Case study

3.2.1. Data set and data pre-processing

The freely accessible data set of the "Battle of Water Demand Forecasting (BWDF)" was utilised, downloaded from https://wdsa-ccwi2024.it/battle-of-water-networks/ (accessed on 25.09.2023). The period up to the first evaluation week (01.01.2021 to 24.07.2022) was used to implement the demand forecasts. This data set includes the net-inflow time series for ten district metered areas (DMAs) with different area characteristics (see also Table 1) on an hourly basis. Additionally, weather data such as air temperature, humidity, precipitation, and wind

Table 1
Overview of district meter area properties.

DMA	Characteristics	Number of users
A	Hospital district	162
B	Residential district in the countryside	531
C	Residential district in the countryside	607
D	Suburban residential / commercial district	2094
E	Residential / commercial district close to the city centre	7955
F	Suburban district including sport facilities and office buildings	1135
G	Residential district close to the city centre	3180
H	City centre district	2901
I	Commercial / industrial district close to the port	425
J	Commercial / industrial district close to the port	776

speed are also part of the data set.

After imputing the missing values using K-Nearest Neighbour (Zanfei et al., 2022b), the time series were resampled to daily values. The average daily water demand over the investigation period varies between 371 and 6765 m^3 (Fig. 1(a)). Some DMAs exhibit significant variations throughout the seasons (e.g., A, B, C, and F) and day of the week (e.g., A, H, I, and J) (Fig. 1(b)).

3.2.2. Input features and forecast approach

A simple feature engineering approach was employed (e.g., without non-linear transformations or principal component analysis) to ensure a comprehensible feature selection process. Additionally, input features and forecasting approaches were selected based on recommendations from literature (Brentan et al., 2017; Kavya et al., 2023). This involved applying a multivariate forecasting model using the time series of water demand itself, calendar inputs, and external weather parameters (precipitation, temperature) as input features. Furthermore, seven models (one for each forecasted day) per ML approach were trained, incorporating the entire development period up to the forecasted day for the external weather parameters. Table 2 summaries the selected input features, which were normalised via z-transformation for training, validation, and testing.

Table 2
Overview of input features for the ML models.

Time	Parameter	Input feature
Historical	Water demand	Daily water demand of last seven days
Future	Calendar input	Month number
		Day of the week (public holidays are categorised as Sundays) of prediction day
	Weather parameters	Daily precipitation sums until prediction day
		Daily mean air temperature until prediction day

3.2.3. Data set splitting

To achieve accurate predictions, the training data must be representative of the entire data set, with particular emphasis on including sufficient peak events. Therefore, the training, validation, and testing splits were carefully defined, using nine splits for hyperparameter tuning on the validation data set to ensure independence from a specific data split. Additionally, the validation and test data were completely independent from the training data used, to estimate how well the model will perform on new and unseen data. As up to seven days of historical water demand values were used as input features, a one-week gap between train and validation, validation and test, and test and train data sets was implemented to ensure no overlap in the data sets. Thus, blocks of three weeks of training data, one week of rest, one week of validation data, one week of rest, one week of test data, and one week of rest were defined, resulting in each split consisting of nine such blocks.

3.2.4. Implementation, hyperparameter tuning and model selection

The Python programming language version 3.9.13 (Python Software Foundation, https://www.python.org/, accessed on 28.03.2024) was employed for implementing all ML models. Specifically, LS, DT, KNN, SVR, and RF were implemented using scikit-learn (Pedregosa et al., 2011). Hyperparameters were tuned via grid-search within the value ranges listed in Table 3. The LSTM model was implemented using Keras (Chollet and others, 2015), and its hyperparameters were tuned using the random search tuner of Keras tune (O'Malley et al., 2019). For each split, the models were trained on the data points defined for training,

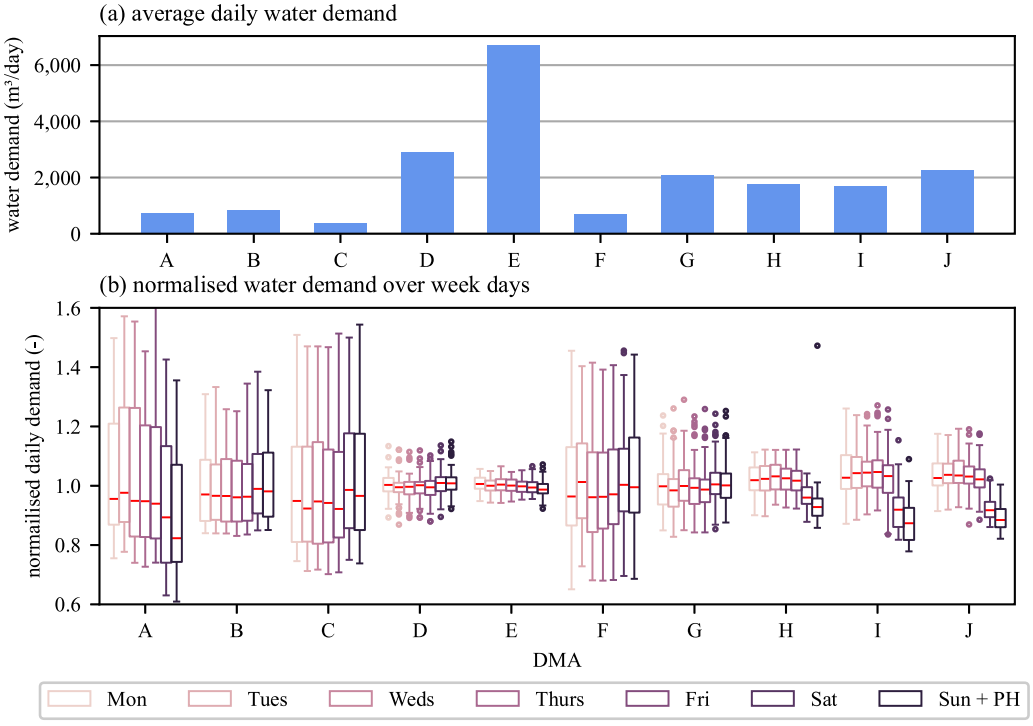


Fig. 1. (a) average daily water demand for the ten DMAs and (b) normalised water demand over weekdays.

Table 3
Overview of applied ML models with tuned hyperparameters.

ML model	Hyperparameter (range for tuning)
LS	–
DT	max_*depth*: np.arange(1, 50), min_*samples_split*: np.arange(5, 20), max_*leaf_nodes*: np.arange(5, 20)
KNN	*n_neighbors*: np.arange(1, 50), *leaf_size*: (20,40,1), *weights*: ['uniform', 'distance']
SVR	*C*: np.logspace(1, 3, 10), *gamma*: np.logspace(-3, 0, 10), *epsilon*: np.logspace(-3,0,10)
RF	max_*depth*: [5,10,15,20], min_*samples_leaf*: [1, 2, 5, 7], *n_estimators*: [10, 20, 50, 70]
LSTM	*batch_size:* [1, 2, 4], *unit_0*: [8, 16, 32], *learning_rate*: [1e-4, 5 × 1e-3]

and the hyperparameters were selected based on validation data. Subsequently, the hyperparameter combination with the lowest error were used for final model training, where the performance was evaluated using test data set to compare the different ML models. Hyperparameters not listed in Table 3 were set to their default values as provided in the aforementioned Python libraries.

In this work, different LSTM architectures were tested. Initially, a bidirectional LSTM structure with two LSTM layers was evaluated as suggested by literature (Zanfei et al., 2022a). Despite varying the number of neurons in both hidden layers and tuning the hyperparameters, satisfactory results could not be achieved. Therefore, the architecture was switched to a unidirectional model with only one hidden layer.

3.3. Evaluation

To assess the accuracy of the predictions, standard metrics as defined in the Python scikit-learn metrics library (Pedregosa et al., 2011) are utilised:

$$\textit{Mean Absolute Error } (MAE) = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{1}$$

$$\textit{Mean Squared Error } (MSE) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{2}$$

$$\textit{Mean Absolute Percentage Error } (MAPE) = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{3}$$

$$\textit{Root Mean Squared Error } (RMSE) = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \tag{4}$$

where y_i *and* $\hat{y}_i$ are the predicted and observed value of the ith sample, respectively, and N is the number of samples. All four metrics are used for model development and evaluation. However, MAPE is employed to measure model accuracy in this work to ensure comparability between DMAs with both low and high total water consumption.

Apart from accuracy, this work focuses on explainability and reliability as outlined in the EU AI Act for Trustworthy AI (European Parliament and Council, 2024) and the EU Ethic Guidelines (European Commission, 2019). To assess these aspects, the applied ML models were first compared regarding their explainability, aiming to address the following questions:

- *Q1: Which input features are used by the model and how do they influence the result?*
- *Q2: How is the prediction influenced by changing parameter values?*
- *Q3: How does the model come to the decision for the forecast on day x?*

For transparent models, such as LS, the questions Q1 and Q3 can be directly retrieved from model parameters (e.g., the regression coefficients). In contrast, achieving explainability for opaque methods is considerably more challenging, requiring post-hoc explainability techniques to address these questions:

- For Q1, SHapley Additive exPlanations (SHAP) is applied, one of the most widely used techniques in XAI research (Belle and Papantonis, 2021). SHAP, based on game theory, calculates SHAP values, which indicate the expected contribution of each individual feature to the model's prediction. Therefore, the marginal contribution of the feature is assessed across all possible permutations of feature subsets (Lundberg and Lee, 2017). This information helps in understanding the impact of each feature on the final prediction.
- To address Q2, partial dependence plots (PDPs) are employed as a visual explanation to illustrate the dependency between input features and predictions. The mean change in outcomes is calculated by varying the target features while holding other features constant. This provides information about the nature of dependency (e.g., linear or nonlinear) and the average effects of the analysed features on the predictions (Belle and Papantonis, 2021).
- For Q3, Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016) is employed, which explains the predictions of any machine learning model by approximating it locally with an interpretable model. LIME constructs a linear surrogate model centred around the predicted value, using the linear regression coefficients to explain predictions made by the LSTM model (Belle and Papantonis, 2021). This allows users to understand the local behaviour of complex models, highlighting which features most strongly influence a particular prediction.

LIME was implemented using the work of Ribeiro et al. (2016), while Alibi explain (Klaise et al., 2021) was utilised for SHAP and PDPs.

Subsequently, the most important input features for the water demand forecasts are selected, and their influence on deviations is investigated to further evaluate the performance of the applied ML models.

4. Results and discussion

4.1. Accuracy

In Fig. 2(a), model accuracy, arguably the most important aspect of technical robustness and safety, is discussed using the forecast for three days ahead as an example. As previously described, the forecasts depend on the last known water consumption values and accurate weather data. The discussion primarily focuses on MAPE, as it is the most interpretable and understandable metric for the reader. The boxplots illustrate the errors of the different approaches on the nine defined splits per DMA. Upon initial inspection, while the errors for all methods appear relatively close to each other across most DMAs, significant differences are observed between individual DMAs.

Referring to Table 1, DMA A, categorised as a "hospital district" and the smallest DMA with only 162 users, exhibited the highest variance in water demand. As depicted in Fig. 2(a), this district also recorded the highest MAPE, with the variance of the errors being notably pronounced, as expected. Particularly, the LSTM model demonstrated varying results depending on the split. DMAs C and F, which showed higher variance in water consumption values, also exhibited higher MAPEs during testing. In contrast, DMA E, the largest district by users described as a residential / commercial district near the city centre with nearly 8000 users, displayed the lowest variance in water demand and the most accurate prediction results.

When forecasting several days in advance (Fig. 2(b), exemplified by DMA C, it is evident that errors increase with the forecast horizon across all tested ML models. Furthermore, the range of errors between individual ML models also widens. For the selected DMA C data set, both SV and RF models stood out of their accuracy. Notable, the RF approach

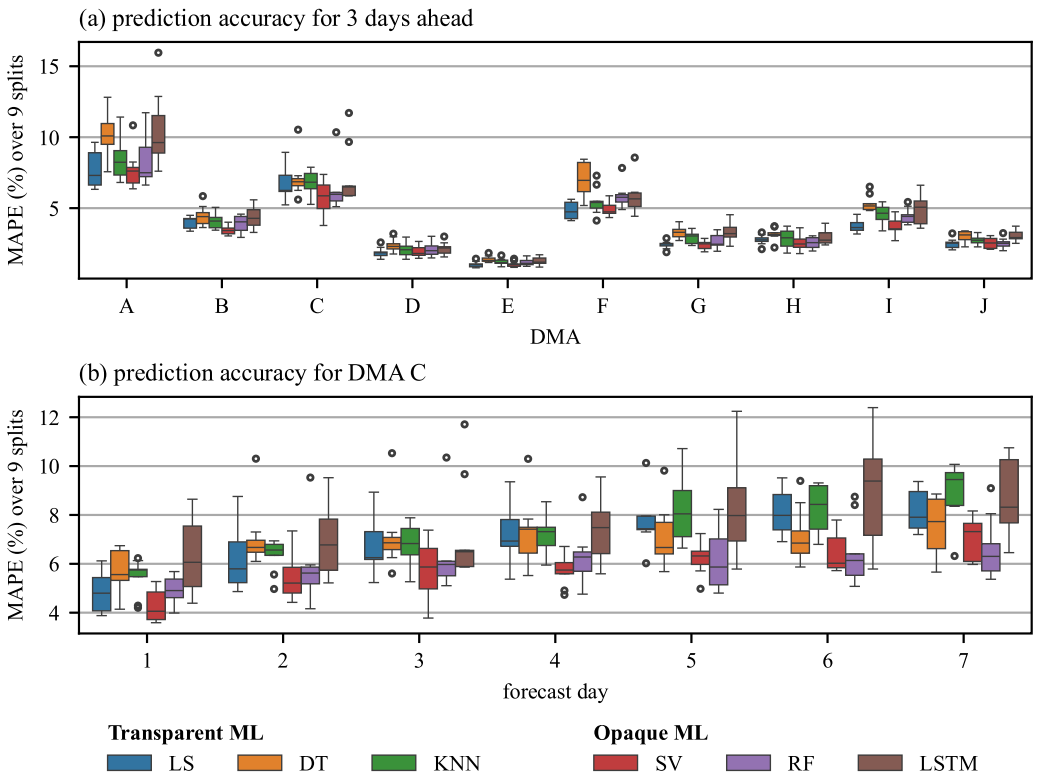


Fig. 2. Exemplary performance comparison for (a) forecast day 3 for all DMAs and (b) for all forecast days in DMA C over the nine data splits.

could potentially be enhanced further using boosting techniques such as XGBoost (Niazkar et al., 2024). Surprisingly, despite efforts to optimise the architecture and hyperparameters, the LSTM model's performance was relatively low, contrary to current literature findings (Kavya et al., 2023). Additionally, it has also the highest variations between the different data splits. This discrepancy could be due that tuning of the hyperparameters, optimisation of the network architecture and splitting of the data were not fully exploited. However, this was deliberately not done to keep the effort per model implementation roughly the same for comparison. Particularly, the approach utilised for splitting the time series into training, validation and test data sets was not optimal for the memory of the LSTM network but was maintained to ensure comparability of the individual ML models. Subsequently, addressing these aspects offer potential to increase the LSTM model's performance as achieved by literature."

Similar results for all investigated metrics and DMAs can be found in the appendix. Key observations include:

- Generally, smaller and more inhomogeneous DMAs exhibited higher variance in water consumption and greater model errors and scattering in their predictions, and vice versa. LS performed exceptionally well across diverse data sets, highlighting its robustness to outliers and noisy data.
- Homogeneous data set showed less variation in error rates among different methodological approaches.
- RF consistently outperformed DT across all DMAs and forecast horizons.
- LS models remained competitive for short-term forecasting (up to three days in advance) across all DMAs, outperforming neural networks like LSTM in this work.
- SV and RF demonstrated superior performance for DMA C, particularly in multiday forecasting scenarios. DT, KNN and LSTM appeared less suitable in this work.

In conclusion, the most suitable approach in terms of model accuracy depends on DMA characteristics, data set, and forecast horizon. As previously discussed, while accuracy is crucial for technical robustness, it must be carefully balanced with other factors such as model transparency in model selection, tailored to specific applications and processes.

4.2. *Explainability*

Model explainability is exemplary discussed for DMA C (split five, forecast day three). This configuration is selected because DMA C has relatively high fluctuations in daily water demand (Fig. 1) and errors in accuracy (Fig. 2(a)). Additionally, there are differences in the accuracy of the individual methods, however, there are no irregularities caused by individual large consumers as in DMA A. Thus, this choice enables a thorough analysis and evaluation of the differences between the implemented ML models.

Q1: Which input features are used by the model and how do they influence the result?

Fig. 3 illustrates the feature importance across all ML models used in this work. Despite not being essential, SHAP was also applied to transparent models for better comparability. Thereby, positive SHAP values enhance the prediction, whereas negative values reduce the forecast value. Two general trends in feature importance became apparent: more complex models placed greater emphasis on daily weather forecasts, whereas simpler models relied heavily on historical water demand. For instance, in the LSTM model, the air temperature on the day of forecast (temp-1) emerged as the most important feature, with higher temperatures correlating with increased water demand predictions. Additionally, (perfectly forecasted) precipitation on the forecast day (rain-1) and of the previous day (rain-0) significantly influenced demand predictions, reducing the forecasted water demand. In contrast, for the LS model, the most important inputs were the time series itself, particularly water consumption values from previous day (C_v–1) and a week prior (C_v–7). Higher historical water demands led to higher forecast values, while external weather predictions had minimal impact on LS model

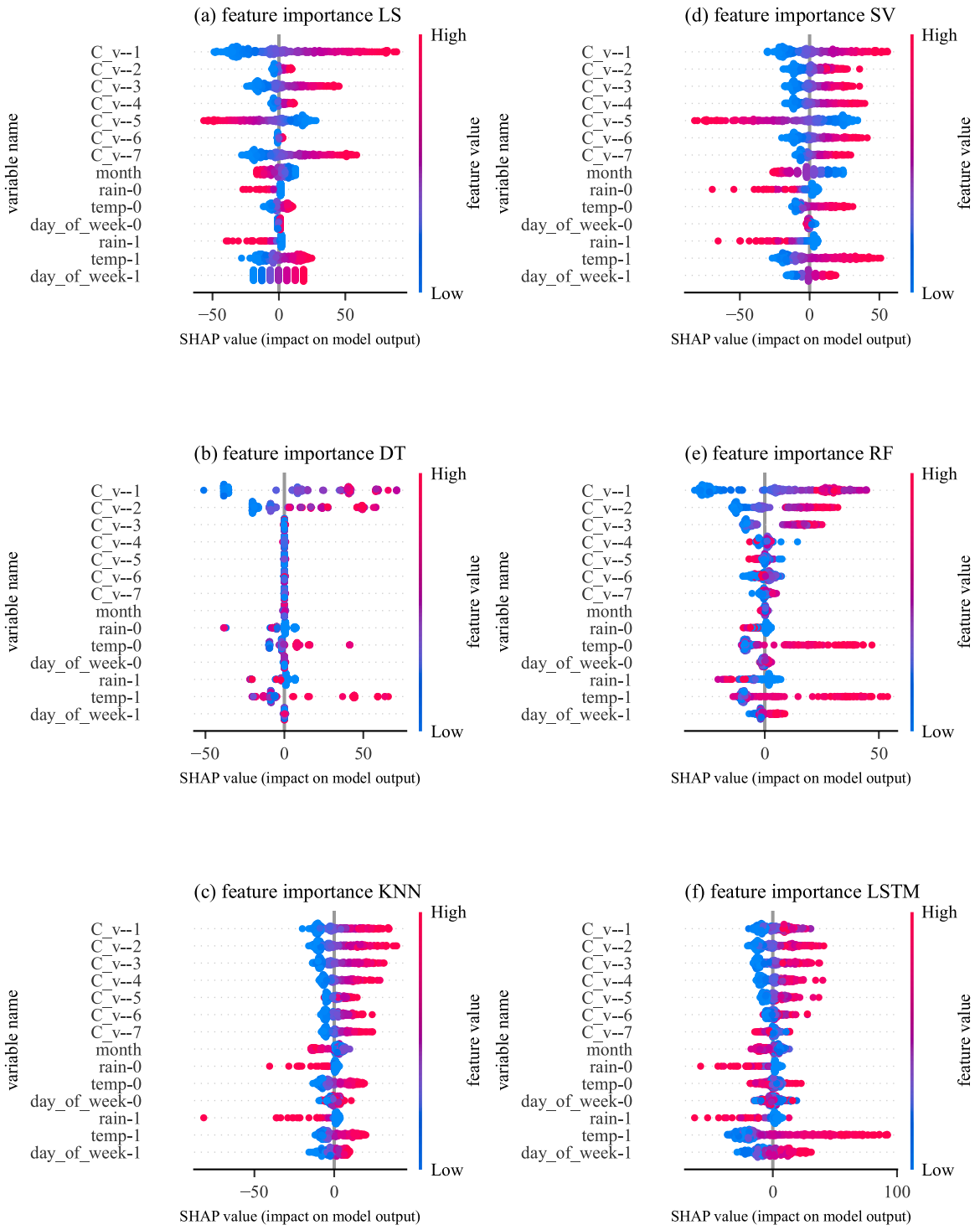


Fig. 3. SHAP plots for (a) LS, (b) DT, (c) KNN, (d) SVM, (e) RF, and (e) LSTM model to determine feature importance.

performance.

In the following discussion, the LS and LSTM models are further employed as exemplars of a transparent (LS being the simplest) and an opaque model (LSTM being the most complex), respectively. It is important to note that the subsequent analyses can and should be conducted for all models tested. The detailed examination of the LSTM model is specifically undertaken to enhance the understanding of the accuracy results (Section 4.1).

Q2: How is the prediction influenced by changing parameter values?

Fig. 4(a) and (b) illustrates the PDPs for previous day's water demand (C_v–1) for the LSTM and LS models, respectively. As suggested by the SHAP values, the previous day's water demand significantly influenced the results of the LS model, whereas it had minimal impact on the LSTM model. For the LSTM model, the average predicted water demand fluctuated between 365 and 407 m^3/day when the previous day's water demand was below 450 m^3/day. Thereafter, it remained constant around 395 m^3/day. In contrast, the PDP for the LS model, showed a linear relationship, reflecting its regression coefficients. Here, the forecasted water demand ranged from 350 to 460 m^3/day as the previous day's demand varied between 300 and 550 m^3/day.

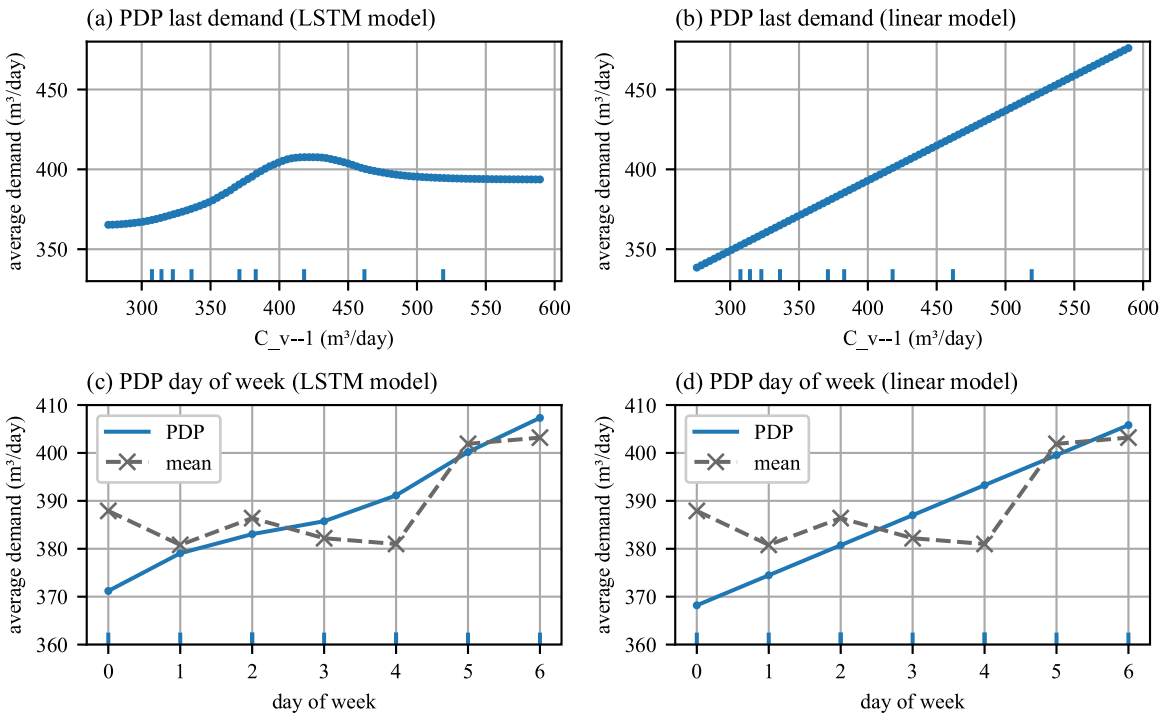


Fig. 4. Partial dependence plots of last known demand for (a) opaque and (b) transparent and for day of week for (c) opaque and (d) transparent model.

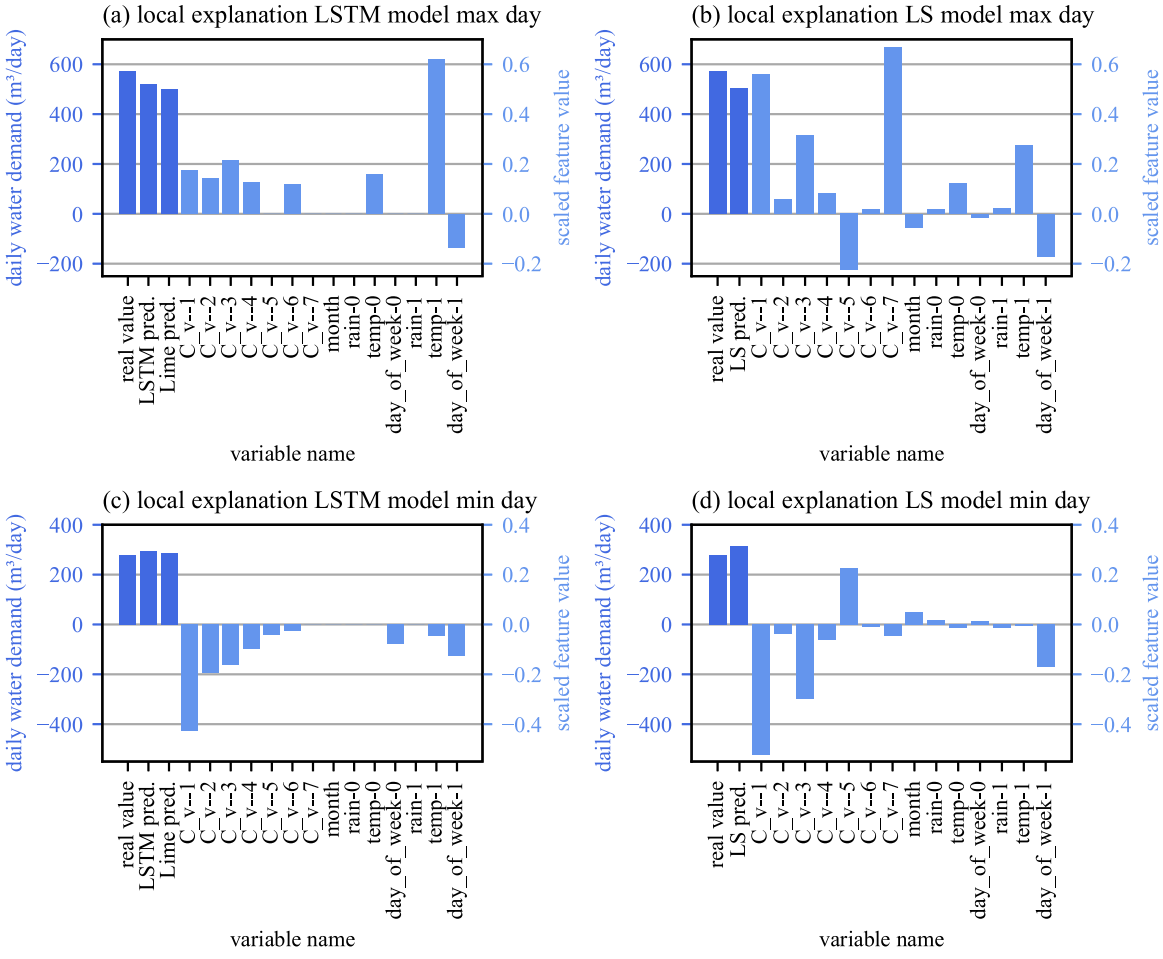


Fig. 5. Local explainability of the maximum water demand for (a) opaque and (b) transparent and for minimum water demand for (c) opaque and (d) transparent model.

Another example illustrates the impact of the day of the week. For the LS model, the predicted water demand averaged around 370 m^3/day on Mondays. It increased steadily to 405 m^3/day by Sundays, indicating a linear progression throughout the week (Fig. 4 (d)). This underscores a significant limitation of the LS model, which can only model linear relationships and cannot capture non-linear fluctuations in consumption across different days of the week. One potential solution to overcome this limitation is to incorporate a one-hot encoded input feature for each day. In contrast, the LSTM model was unaffected by this issue and demonstrated a better capability to model variation in water demand throughout the week (Fig. 4 (c)).

Q3: How does the model come to the decision for the forecast on day x?

To address this question, the detailed decision process is examined for days with the minimum and maximum water demand in the test data set, recorded at 277 and 572 m^3/day, respectively. For the day with maximum water demand (Fig. 5(b)), the LS model predicted 504 m^3/day. The feature 'C_v–7′ had the strongest influence on the prediction, indicated by it scaled feature value of 0.63. Additionally, 'C_v–1′, 'C_v–3′, and 'Temp-1′ positively contributed to the prediction, while 'rain-1′ and 'day-of-the-week' had a negative impact. Similarly, these input features also affected the predictions for the day with minimum water demand, predicting 313 m^3/day. However, the effects were reversed (Fig. 5(d)).

The LSTM model predicted a maximum water demand of 560 m^3, whereas the LIME surrogate model returned a value of 538 m^3/day (Fig. 5(a)). The prediction was primarily influenced by 'temp-1′, with a scaled feature value of 0.54, followed by historical water demand over the last four days. For the day with minimum water demand, the LSTM model predicted 321 m^3/day (Fig. 5(c)). The LIME model predicted 314 m^3/day, with all features exerting a negative influence on the results. Interestingly, historical water demand had the greatest impact on the prediction, suggesting that its influence diminished as temperature decreased, as also indicated by the feature importance in Fig. 3(e)).

4.3. Reliability

The explainability analyses in the preceding section revealed that forecasted values depend significantly on weather forecasts and historical water demand. Therefore, the following aspects are examined to evaluate the reliability of the ML models:

- *Q4: What is the influence of deviations in weather forecasts on the forecast quality?*
- *Q5: What is the influence of irregularities in the water demand data?*

Q4: What is the influence of deviations in weather forecasts on the forecast quality?

For this question, partial dependence plots (PDP) are once again applied to illustrate the combined influence of forecasted precipitation and air temperature on the average change in results. The 2D plots reveal that the LSTM model demonstrated a non-linear relationship between precipitation and temperature (Fig. 6 (a)). Conversely, the LS model, as expected, showed a linear relationship between these two input features (Fig. 6(b)). The predictions of the LSTM model were notably influenced by higher temperatures, consisting with the findings from feature importance (*Q1*) and model explainability (*Q3*). Consequently, accurate weather forecasts are particularly important for high air temperatures, as deviations from predicted weather conditions can significantly affect the predictions. In contrast, the LS model was generally less sensitive to weather parameters. As a result, weather forecast deviations had a lesser impact on its prediction values.

Q5: What is the influence of irregularities in the water demand data?

As shown in Fig. 7(a), the daily water demand during winter 2021 was higher than in winter 2022, possibly influenced by repaired leakages, changes in large water consumers, or work regulations during the Covid-19 pandemic. Since water demand served both as input features and target value, it significantly impacted the model's quality and predictions. To assess the ML models' sensitivity to these changes, the difference between the minimum water demand in winter 2022 and 2021 is subtracted for two periods: from 01.01.2021 to 25.05.2021 (before summer demand, referred to as variant 1) and from 01.01.2021 to 16.09.2021 (after summer demand, referred to as variant 2). Afterwards, a LSTM and LS model were trained for each modified data set.

Fig. 7(b) and Fig. 7(c) show the prediction results for the last week in the test data for the LSTM and LS models, respectively. These results were compared with the real water demand and predictions using the original ML models. Surprisingly, the LS model showed minimal sensitivity to changes in the training data, despite historical water demand being its most important input feature. It predicted an identical pattern, and all forecast values are 0.5 m^3/day and 30 m^3/day lower compared to the original predictions for variant 1 and variant 2, respectively. In contrast, the modified training data significantly affected the prediction results of the LSTM model, altering both the weekly pattern and increasing or decreasing the predicted values. The maximum differences are 30 m^3/day for variant 1 and 80 m^3/day for variant 2 compared to the forecast values of the original time series.

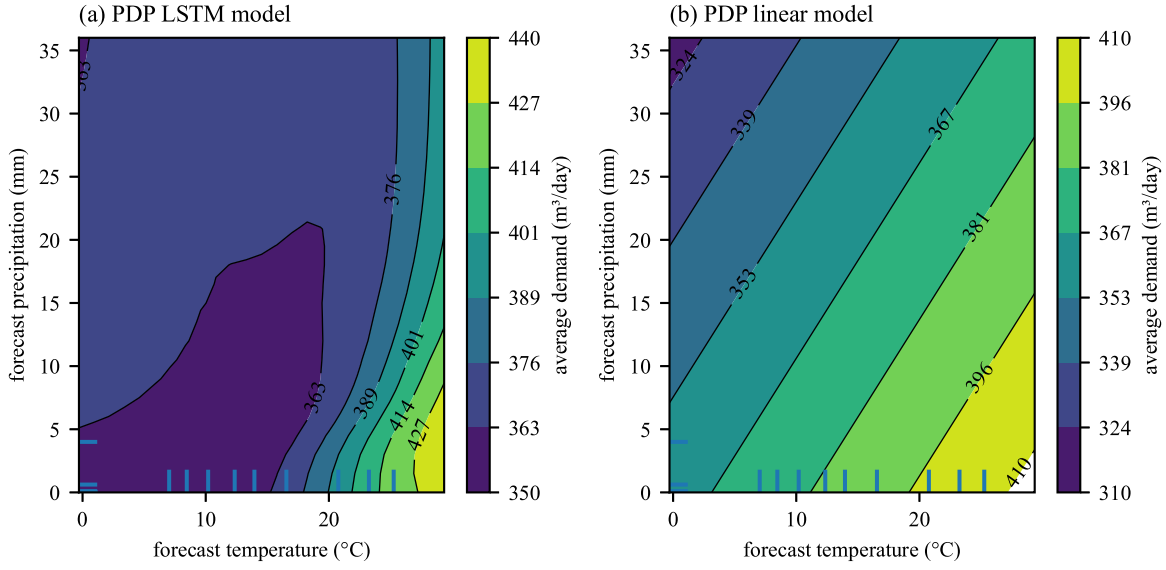


Fig. 6. Partial dependence plots for (a) opaque and (b) transparent model to determine influence of weather forecasts on the average predicted water demand.

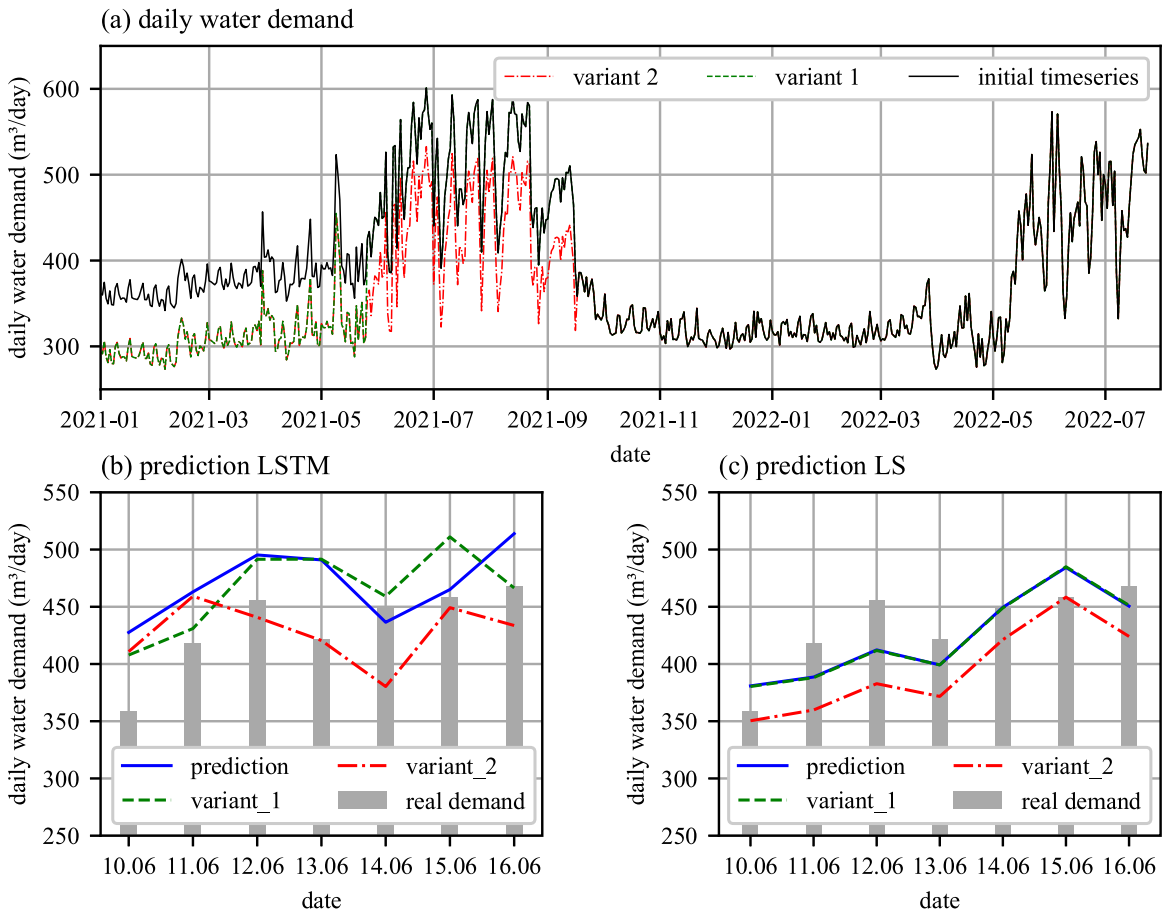


Fig. 7. (a) Measured water demand with different level of corrected input data and daily predictions of the last week of the test data set for (b) the LSTM and (c) the LS model (the prediction line is hidden by variant_1, as the difference is only 0.5 m^3/day).

4.4. Recommendations

As stated in the EU AI Act, model selection should undergo a risk assessment, weighting accuracy against model explainability and reliability. The achievable prediction accuracy largely depends on the characteristics of the DMA, with both transparent and opaque models demonstrating good performance.

However, opaque models generally achieve better prediction accuracy due to their capability to analyse more complex data and nonlinear relationships. As highlighted in the explainability analysis, future weather data emerged as the most important input feature for forecasting future water demand with opaque models. As opaque models aim to capture the complex relationship between weather data and water demand, they need sufficient quantity and quality of historical water demand data and weather data to accurately capture all relationships and prevent modelling wrong correlations. The LSTM model, being the most complex ML studied, is particularly sensitive to these requirements. This sensitivity is indicated by the explainability and reliability analysis, which explains its poorer accuracy compared to other ML models. Consequently, testing a different categorisation of the training, validation, and testing data (without a one-week rest and with an equal distribution of low, medium, and peak values) significantly enhanced the LSTM model's performance. Moreover, accurate weather forecasts are essential for opaque models, as deviations greatly impact prediction values.

In contrast, transparent models typically exhibited lower accuracy compared to opaque models due to their lower complexity. However, they rely heavily on historical water demand data as the most important input features. This reliance enhances their robustness against variations in the quantity and quality of training data, as well as deviations in the weather forecasts.

According to the AI Act, the investigated use case of water demand forecasting for tank operations can be classified as high-risk AI. As discussed throughout this work, both transparent and opaque models meet the requirements for accuracy, explainability, and reliability when appropriately applied. The choice of model also depends on the operator's preferences. The authors recommendation would be to use transparent models for the tested DMAs. These models are less influenced by weather data forecasts, which intrinsically contain unavoidable uncertainties. Particularly, linear regression models are inherently robust against various data irregularities, are easy to implement, and require no hyperparameter tuning, while still providing predictions in sufficient quality. Additionally, linear models can be extended with Bayes statistics to Bayesian Linear Regression to provide an uncertainty assessment, enhancing their capability to predict peak demands accurately (Taheri and Razban, 2022).

5. Summary and conclusions

The European parliament and the Council approved the "Artificial Intelligence Act" in March 2024, establishing a comprehensive legal framework for artificial intelligence (AI). This framework identifies high-risk AI applications, such as those in the water supply system (WSS), which must adhere to stringent requirements. The first part of this work involved a legal assessment of the AI Act to evaluate its impact on AI applications within the WSS. Subsequently, the regulatory requirements were applied to a specific use case: water demand forecasting aimed at enhancing daily system operations and management.

Six standard machine learning (ML) models were selected for systematic comparison: Linear Regression, Decision Tree, and K-Nearest Neighbours as transparent models, and Support Vector Machine, Random Forest, and Long Short-Term Memory as opaque models. This comparison utilised the open-data set from the "Battle of Water Demand Forecasting (BWDF)" (https://wdsa-ccwi2024.it/battle-of-water-networks/, accessed on 25.09.2023). Beyond accuracy, this work focused on the explainability and reliability of these ML models to understand their decision-making process and identify potential limitations.

Based on the obtained results, the following conclusions can be drawn:

- AI applications in WSS: AI applications are not inherently categorised as high-risk AI under the AI Act. High-risk AI refers to implementations, where failures or lack of transparency results can significantly impact physical infrastructure or supply reliability.
- Water demand forecasts as high-risk AI: Water demand forecasts for controlling tank operations are classified as high-risk AI, for example. Errors in these forecasts could lead to insufficient availability of drinking water. High-risk AI applications must meet special requirements for model transparency (traceability, explainability, communication) and technical robustness (amongst others accuracy and reliability).
- Performance of opaque models: Opaque models generally achieve the highest accuracy for the tested DMAs, relying heavily on weather data as most important input feature. They require training data of sufficient quantity and quality to accurately represent the relationship between weather and water demand data. However, they are more vulnerable to deviations and irregularities in weather forecasts than the transparent models.
- Performance of transparent models: For the tested data sets, transparent models generally have lower prediction accuracy compared to opaque models. They mainly rely on historical water demand data and are less influenced by weather data, making them more robust against various data irregularities.
- Model selection considerations: Both transparent and opaque models meet the requirements for explainability. The choice of model should also consider the operator's preferences and the context of the application.

CRediT authorship contribution statement

Claudia Maußner: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Martin Oberascher:** Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Arnold Autengruber:** Writing – original draft, Methodology, Investigation. **Arno Kahl:** Writing – review & editing, Supervision, Methodology. **Robert Sitzenfrei:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This publication was produced as part of the "REWADIG" project. This project is funded by the Climate and Energy Fund and is part of the programme "Smart Cities Demo – Boosting Urban Innovation 2020″ (project 884788).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2024.122779.

Data availability

The freely accessible data set of the "Battle of Water Demand Forecasting (BWDF)" was utilised.

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